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**VISION-BASED REAL-TIME SCREEN LAYOUT ANALYSIS
AND BACKGROUND ANOMALY DETECTION**

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfillment for the Course of

ITA0512 – COMPUTER VISION

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Vision-Based Real-Time Screen Layout Analysis and Background Anomaly Detection**” has been carried out by **Galla Sai Siddhartha** under the supervision of **Dr. Christy Melwyn** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech **Artificial Intelligence & Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **Galla Sai Siddartha** of the **B.Tech Artificial Intelligence & Machine Learning**, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Vision-Based Real-Time Screen Layout Analysis and Background Anomaly Detection**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Galla Sai Siddartha (192425093)	Prepared the screen-capture and dual-OCR workflow for Module 1; defined text, region, and confidence fields for downstream analysis..	Integrated capture preprocessing, OCR-A/OCR-B fusion, and layout evidence extraction.	Checked OCR confidence, missed text cases, scaling effects, and false text detections.	Prepared the Module 1 supporting sections, extraction diagrams, and related appendix evidence.	33%
Kandukuri Vijay Bhaskar Reddy (192425155)	Prepared the object-detection dataset structure and YOLOv8 inference configuration.	Integrated YOLOv8 bounding boxes, classes, confidence values, and frame-level detections.	Tested precision, recall, IoU thresholds, and detection behaviour under screen changes.	Prepared the object-detection and evaluation sections of the report.	34%
Yuvaraj D (192421230)	Designed the risk-scoring features and background-anomaly decision logic.	Implemented feature weighting, thresholds, event states, and audit-record fields.	Analysed threshold sensitivity, false positives, false negatives, and latency.	Prepared the risk, security, and technical evaluation documentation.	33%
Total					100%

ABSTRACT

Modern computer users interact with complex screen layouts containing application windows, text panels, overlays, status indicators, notifications, and background processes. A system that can analyse these layouts in real time can support accessibility, operational monitoring, security review, and safer human–computer interaction. This project presents a Vision-Based Real-Time Screen Layout Analysis and Background Anomaly Detection system for capturing screen frames, extracting visible text through a dual OCR pipeline, identifying relevant objects and regions with YOLOv8, assigning a risk score to unusual visual or background behaviour, and generating controlled alerts. Module 1, Screen Capture & Dual OCR, is included as a supporting capability where necessary because reliable text and layout evidence improves the downstream risk decision; the report does not treat Module 1 as a complete standalone implementation chapter. The engineering design separates capture, preprocessing, OCR, object detection, risk scoring, policy enforcement, alert delivery, and audit logging. It emphasizes confidence thresholds, human review, reversible actions, privacy-aware retention, and safe failure. The proposed system is evaluated through OCR quality, layout localisation, detection precision and recall, risk-classification scenarios, alert task completion, response latency, and comparison with a simpler baseline. The intended contribution is a maintainable computer-vision prototype that transforms screen observations into explainable events without claiming that a visual anomaly alone proves malicious intent.

Keywords: computer vision, screen capture, optical character recognition, OCR, YOLOv8, object detection, anomaly detection, risk scoring, real-time monitoring, alerts

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	1
2	CHAPTER 2: Literature Review	4
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	6
4	CHAPTER 4: Implementation, Testing & Results, Project Management	12
5	CHAPTER 5: Conclusion & Reflection	16
	References	20
	Appendices	21

LIST OF FIGURES

S.No.	Figure title	Page No
1	Figure 1. Platform Context and Architecture	3
2	Figure 2. Screen Capture and Dual OCR Flow	8
3	Figure 3. Natural-Language and Layout Evidence Extraction	9
4	Figure 4. YOLOv8 Detection and Risk Scoring Path	10
5	Figure 5. Active Enforcement and Alerts	11
6	Figure 6. Illustrative Screen Monitoring Dashboard	14
7	Figure 7. Illustrative Evaluation Comparison	14

LIST OF TABLES

S.No.	Table title	Page No
1	Table 1. Comparative Analysis of Existing Approaches	4
2	Table 2. Stakeholder and User Requirements	6
3	Table 3. Functional and Non-Functional Requirements	6
4	Table 4. Constraints and Applicable Standards	7
5	Table 5. Design Specifications	8
6	Table 6. Alternative Solutions Evaluation	8
7	Table 7. Sustainability, Ethics, Safety and Security Assessment	9
8	Table 8. Risk Register	10
9	Table 9. Implementation Tools and Resources	11
10	Table 10. Test Plan and Verification	12
11	Table 11. Achievement of Objectives	14
12	Table 12. Roles, Milestones and Budget	15

LIST OF ABBREVIATIONS AND SYMBOLS

Abbreviation/Symbol	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CV	Computer Vision
FPS	Frames per second
IoU	Intersection over Union
OCR	Optical Character Recognition
ROI	Region of Interest
YOLO	You Only Look Once
TP / FP / FN	True positive / False positive / False negative
PII	Personally Identifiable Information

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Screen layouts are visual environments rather than ordinary image collections. They contain text, windows, controls, icons, pop-ups, status messages, and background regions that can change from frame to frame. In many practical settings the relevant question is not merely which object is present, but whether the composition of the screen is expected for the current task. For example, a training workstation may show a coding editor, a terminal, a browser, and a project dashboard. A newly appearing overlay, unknown window, or background process indicator may require attention even when no single object is obviously dangerous.

Computer vision can support this analysis by combining screen capture, image preprocessing, OCR, object detection, temporal comparison, and event classification. OCR converts visible text into searchable evidence. A detector such as YOLOv8 returns bounding boxes, class labels, and confidence scores for objects or regions of interest. A risk engine can combine these observations with layout history, allow-lists, and policy rules. The result is an evidence-based event rather than a conclusion based on a single pixel pattern.

1.2 Need for the Project

Manual screen inspection is slow and difficult to reproduce. Conventional screenshot utilities can capture an image but do not explain the layout or compare background changes. OCR-only systems provide text but may miss the relationship between a message and its surrounding window. Object-detection-only systems identify visual regions but may not understand the text displayed inside them. The project is needed to join these evidence types in a controlled real-time pipeline.

1. The system should observe screen layout changes without requiring constant manual inspection.
2. Text should be extracted with confidence values and region coordinates so that uncertain results can be reviewed.
3. Visual detections should be combined with temporal and policy features rather than interpreted in isolation.
4. Alerts should be proportionate, reversible, and recorded in an audit log.

5. The design should minimise retention of sensitive screen content and provide clear user consent and access controls.

1.3 Problem Statement

The engineering problem is to design and evaluate a real-time computer-vision service that captures screen frames, extracts text and layout evidence, detects relevant objects or anomalies, calculates a transparent risk score, and delivers policy-controlled alerts. The system must balance detection sensitivity with false-alarm cost, real-time performance with computational resources, and observability with privacy. It must also handle incomplete OCR, changing screen resolutions, overlapping windows, transient notifications, and benign software updates.

1.4 Aim and Objectives

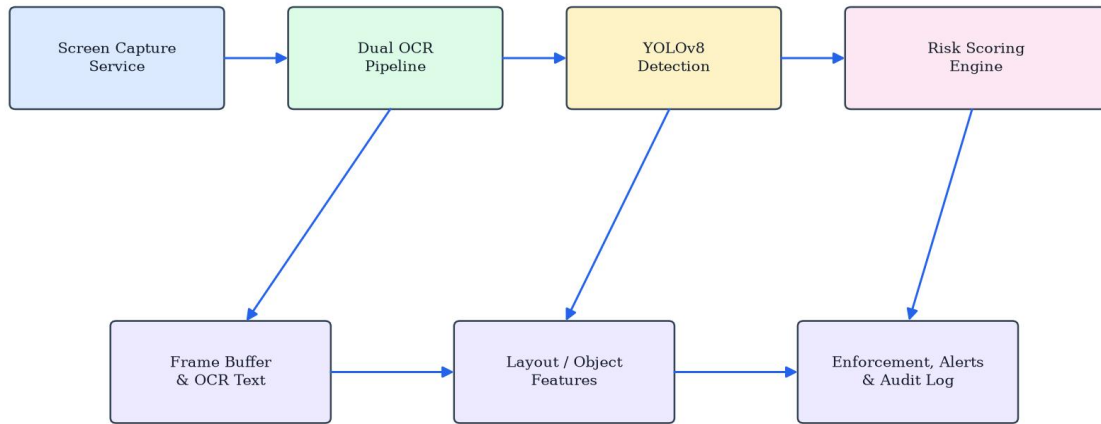
The aim is to develop a responsible vision-based prototype for real-time screen layout analysis and background anomaly detection. The objectives are:

1. To capture screen frames at a configurable rate and preprocess them for analysis.
2. To implement a dual OCR workflow that combines full-frame text extraction with region-based OCR and confidence recording.
3. To integrate YOLOv8 detections with coordinates, class labels, confidence, and temporal evidence.
4. To calculate an explainable risk score using visual, textual, layout, and temporal features.
5. To implement active enforcement and alerts with soft-block, acknowledgement, escalation, and audit states.
6. To evaluate quality, latency, false-positive behaviour, and usability using reproducible scenarios.

1.5 Scope and Expected Outcome

The scope includes desktop or virtual-screen capture, OCR, layout analysis, YOLOv8 inference, risk scoring, alert workflows, audit records, and evaluation. Module 1 is treated as a supporting module: its outputs are used where necessary to improve text and layout evidence, but this report does not repeat every possible OCR technique. The project does not claim to determine user intent, prove malware, conduct covert surveillance, or replace endpoint-security controls. The expected outcome is a demonstrable prototype, technical report, evaluation dataset or scenario set, and evidence of responsible operation.

Vision-Based Real-Time Screen Layout Analysis and Background Anomaly Detection



Controlled pipeline: capture → OCR → detection → risk decision → alert or audit.

Figure 1. Platform Context and Architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review used a focused engineering approach covering computer-vision libraries, OCR systems, object-detection documentation, anomaly-detection methods, screen-monitoring patterns, secure software design, accessibility, and responsible AI guidance. Sources were chosen when they informed technical feasibility, evaluation practice, privacy, safety, or risk management. The review supports design decisions rather than claiming a systematic meta-analysis.

2.2 Existing Technical Approaches

OpenCV provides a widely used foundation for image capture and preprocessing. OCR engines such as Tesseract and neural OCR pipelines can identify text, but their performance depends on scale, contrast, language, and layout. Ultralytics documentation describes object detection outputs as bounding boxes, class labels, and confidence scores, which are suitable inputs for a downstream risk engine. Modern detection models can be accurate and fast, but deployment still requires threshold selection, calibration, and attention to domain shift.

Screen monitoring systems may also rely on operating-system events, application allow-lists, or security telemetry. These sources can provide useful context but do not always describe what is visually displayed. A hybrid approach is therefore appropriate: vision provides observed screen evidence, while policy and system context reduce ambiguous interpretation.

2.3 Comparative Analysis

Table 1. Comparative Analysis of Existing Approaches

Past / reference approach	Advantages	Limitations	Relevance to proposed work
Manual screenshot inspection	No model training; human can interpret context.	Slow, inconsistent, and difficult to run continuously.	Motivates automated observation with human review.
Screen capture without analysis	Simple and low cost; preserves visual evidence.	Does not identify text, layout, objects, or risk.	Provides the input stream for the proposed pipeline.

OCR-only monitoring	Can search visible text and status messages.	Weak on visual context, regions, overlays, and non-text anomalies.	Used as a component through dual OCR evidence.
Object-detection-only monitoring	Provides boxes, classes, and confidence values.	May not understand displayed text or temporal policy context.	Used as the YOLOv8 evidence layer.
Rule-based allow-list systems	Transparent and easy to audit.	Rigid rules create false positives and miss unseen patterns.	Combined with learned visual evidence and thresholds.
Proposed hybrid platform	Combines capture, OCR, YOLOv8, temporal features, risk scoring, alerts, and audit.	Requires calibration, privacy controls, and testing against benign variation.	Addresses the identified integration gap.

2.4 Engineering Gap and Proposed Contribution

The gap is not the absence of an image detector or OCR engine. The gap is a controlled workflow that relates text, layout, visual objects, temporal changes, and policy state in real time. The proposed contribution is a modular design in which Module 1 supplies verified text and region evidence; Module 2 detects objects and calculates risk; and Module 3 manages proportionate actions, alerts, and auditability. The design explicitly separates observation from decision and decision from enforcement.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholders and Requirements

Stakeholders include the monitored user, system administrator, security or operations coordinator, software developer, and institution responsible for data governance. Requirements were derived from the need to observe screen layouts while avoiding unjustified conclusions or excessive retention.

Table 2. Stakeholder and User Requirements

Stakeholder	Requirement	Evidence of satisfaction
Monitored user	Receive clear notice, understandable alerts, and a way to acknowledge or contest an event.	Consent/notice record and feedback workflow.
Coordinator	Review evidence, risk factors, thresholds, and history before escalation.	Review screen and audit event.
Administrator	Configure capture rate, allow-lists, retention, roles, and model versions.	Configuration record and access test.
Developer	Run reproducible inference and evaluation with versioned code and data.	Repository, environment file, and test report.
Institution	Apply privacy, accessibility, security, and incident-response requirements.	Policy checklist and approval evidence.

Table 3. Functional and Non-Functional Requirements

Requirement	Type	Measurable target
Screen capture	Functional	Capture configurable frames without corrupting timestamps.
Dual OCR	Functional	Produce text, regions, confidence, and source-path fields for review.
YOLOv8 inference	Functional	Return class, confidence, and bounding-box coordinates.

Risk scoring	Functional	Display component factors and final risk band.
Alerts	Functional	Deliver a policy-controlled alert and record acknowledgement.
Latency	Non-functional	Report capture-to-decision latency under the selected pilot hardware.
Privacy	Non-functional	Restrict screen images and personal data by role and retention policy.
Maintainability	Non-functional	Version model, threshold, taxonomy, and configuration changes.

Table 4. Constraints and Applicable Standards

Constraint / standard area	Design implication
Technical	Resolution, FPS, OCR language, GPU/CPU availability, and screen scaling affect accuracy and latency.
Economic	Prefer open-source tools and a configurable sampling rate for pilot deployment.
Environmental	Avoid unnecessary full-resolution processing; use region-of-interest and adaptive sampling where safe.
Ethical	Provide consent, transparency, contestability, and a human review path.
Safety	Use soft enforcement before blocking; treat uncertainty as a review condition.
Security	Encrypt stored evidence, apply least privilege, validate inputs, and keep audit logs.
Accessibility	Use readable alert text, colour-independent status, keyboard support, and alternative descriptions.

3.2 Design Specifications

The platform is designed as a staged pipeline. Screen capture creates timestamped frames. Preprocessing normalises scale and contrast. Module 1 runs two OCR paths: a full-frame path for broad coverage and a region path for text boxes or candidate windows. The outputs are fused with confidence and coordinates. Module 2 receives OCR evidence, frame differences, YOLOv8 detections, allow-list context, and temporal features. The risk engine returns a score, band,

explanation, and recommended action. Module 3 applies policy: notify, request confirmation, soft block, escalate, or log for later review.

Table 5. Design Specifications

Component	Input	Output	Design choice
Capture service	Screen or virtual-display frame	Timestamped image and metadata	Configurable sampling with privacy-aware retention.
Dual OCR	Full frame and candidate regions	Text spans, boxes, confidence, source path	Redundant evidence reduces dependence on one OCR path.
YOLOv8 detector	Preprocessed frame	Boxes, labels, confidence	Fast object/region detection with threshold calibration.
Risk engine	Vision, OCR, temporal, policy features	Score, band, explanation	Weighted and rule-aware rather than opaque.
Enforcement service	Risk event and policy	Alert, acknowledgement, escalation, audit	Reversible actions and human confirmation.
Audit store	Events and decisions	History and reports	Versioned, access-controlled, retention-limited records.

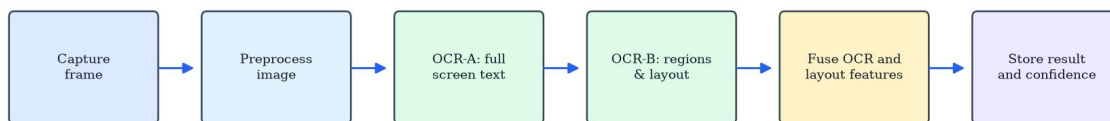


Figure 2. Screen Capture and Dual OCR Flow

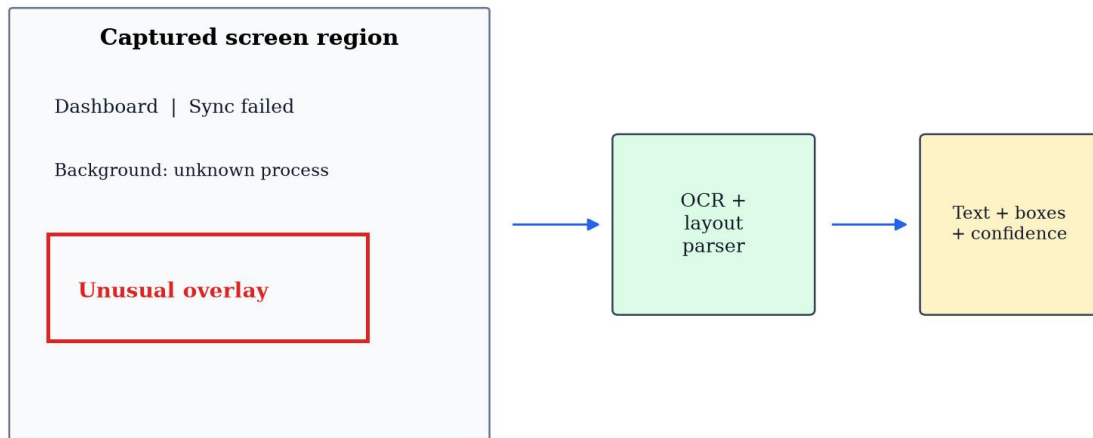


Figure 3. Natural-Language and Layout Evidence Extraction

3.3 Alternative Solutions and Trade-offs

Several alternatives were considered. A pure OCR solution would be easy to explain but would miss visual anomalies. A pure object detector would provide boxes but not the text or semantic context inside windows. A large multimodal model might provide richer descriptions, but it can increase cost, latency, privacy exposure, and difficulty of deterministic evaluation. The selected architecture uses smaller specialised components with explicit fusion and policy logic.

Table 6. Alternative Solutions Evaluation

Alternative	Strength	Weakness	Decision
OCR-only pipeline	Low complexity and searchable text.	Cannot reliably detect non-text visual changes.	Rejected as complete solution; retained as component.
YOLO-only pipeline	Fast boxes and classes.	Limited semantic understanding of screen text and policy context.	Rejected as complete solution; retained as component.
Cloud multimodal model	Flexible descriptions and broad capabilities.	Privacy, cost, latency, reproducibility, and dependency concerns.	Not selected for the core pilot.
Hybrid local pipeline	Explicit evidence, modular testing, and data control.	Requires integration and calibration effort.	Selected solution.

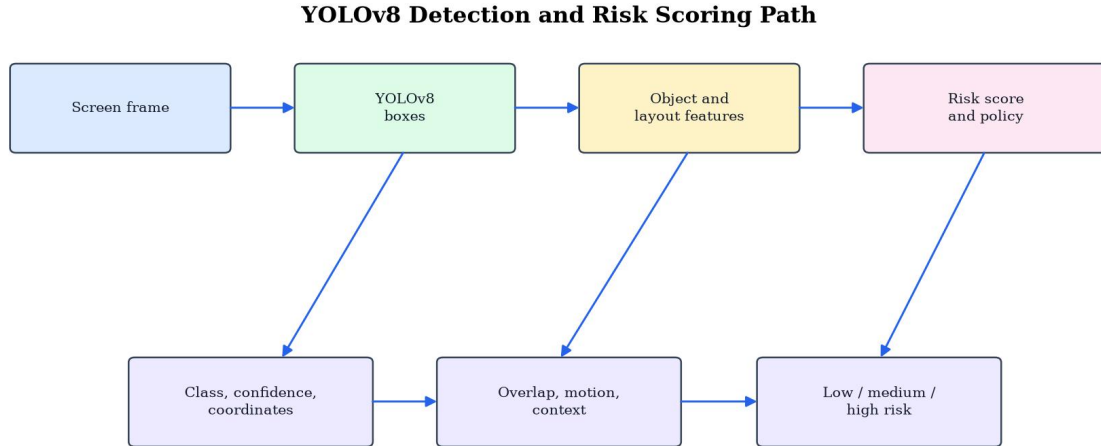


Figure 4. YOLOv8 Detection and Risk Scoring Path

3.4 Sustainability, Ethics, Safety, Risk and Security

The system should process only the screen regions and time windows necessary for the stated purpose. Where possible, the design can store extracted features and event thumbnails instead of continuous raw screen video. Users should know what is captured, how long it is retained, and who may review it. A visual anomaly is an observation, not proof of wrongdoing. Alerts therefore include evidence, confidence, risk factors, and a review route.

Table 7. Sustainability, Ethics, Safety and Security Assessment

Area	Risk or responsibility	Control / mitigation
Sustainability	Continuous high-resolution inference consumes compute and energy.	Adaptive sampling, region-of-interest processing, batching, and model-size selection.
Privacy	Screens may contain credentials, messages, or personal data.	Consent, masking, encryption, role-based access, and retention limits.
Ethics	False alerts may unfairly affect a user.	Human review, explanation, contestability, and reversible enforcement.
Safety	Aggressive blocking can interrupt legitimate work.	Soft block and confirmation before stronger action.
Security	Captured evidence may become a sensitive target.	Least privilege, secure storage, input validation, and audit logs.
Bias and robustness	Model behaviour may vary by theme, font, language, or	Representative test cases, error analysis, calibration, and

	resolution.	monitoring.
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Table 8. Risk Register

Risk	Likelihood	Impact	Mitigation	Owner
OCR misses important text	Medium	Medium	Dual OCR, confidence threshold, manual review.	Module 1 owner
False object detection	Medium	High	IoU/confidence calibration and temporal confirmation.	Detection owner
Benign popup triggers alert	High	Medium	Allow-list, persistence rule, acknowledgement workflow.	Risk owner
Sensitive screen capture retained	Medium	High	Masking, encryption, retention deletion, role restrictions.	Administrator
Alert service unavailable	Low	High	Local queue, retry, audit, and fallback notification.	Enforcement owner
Model drift after UI change	Medium	Medium	Version checks, representative regression set, recalibration.	Project team

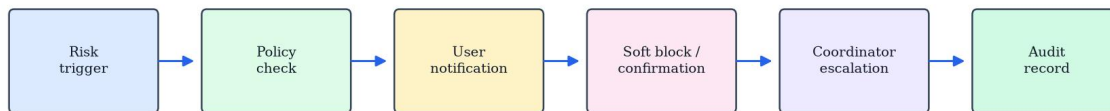


Figure 5. Active Enforcement and Alerts

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Implementation Approach

Implementation is organised into services that can be tested independently. The capture service obtains frames and records metadata. The preprocessing service converts images into consistent dimensions and colour representation. Module 1 extracts text and layout evidence only where required. The detector service runs YOLOv8 inference. The risk service computes features and applies thresholds. The enforcement service sends alerts and records actions. A dashboard presents current state and recent events.

Table 9. Implementation Tools and Resources

Tool / technology	Use in implementation	Reason for selection
Python	Orchestration, preprocessing, evaluation, and services.	Accessible ecosystem for computer vision and experimentation.
OpenCV	Capture integration, resizing, thresholding, and image operations.	Mature computer-vision primitives.
Tesseract or neural OCR	Full-frame and region OCR paths.	Provides text evidence with confidence and boxes.
Ultralytics YOLOv8	Object and region detection.	Practical training, inference, and export workflow.
NumPy / Pandas	Feature processing and evaluation summaries.	Convenient numerical and tabular analysis.
FastAPI or equivalent	Internal API for inference and alerts.	Clear service boundary and testable endpoints.
SQLite/PostgreSQL	Events, configurations, decisions, and audit records.	Structured storage and queryability.

4.2 Module 1 – Screen Capture & Dual OCR: Necessary Coverage

Module 1 is included as a supporting part of the overall design. The project uses it to provide text and region evidence when the risk engine needs to distinguish a normal interface from an unexpected overlay, changed status message, or unknown window. The module captures a frame,

creates a full-frame OCR result, identifies candidate regions, runs region OCR, normalises text, and stores confidence with the source path. Low confidence does not directly trigger enforcement; it requests review or causes the risk score to be reduced.

A dual path is useful because full-frame OCR provides coverage while region OCR can improve small or localised text. The two outputs are reconciled by comparing text similarity, region overlap, and confidence. The fused record contains the original frame identifier, OCR source, text span, bounding box, confidence, and timestamp. The remainder of the report focuses on the integrated system rather than describing every OCR algorithm in detail.

4.3 Testing and Verification

Testing combines unit tests, integration tests, scenario tests, performance measurements, and human review. OCR tests use known text at multiple scales and themes. Detection tests use labelled screen regions and measure precision, recall, IoU, and mAP where appropriate. Risk tests verify that score components are visible and that hard policy conditions are respected. Enforcement tests verify acknowledgement, escalation, reversal, and audit records.

Table 10. Test Plan and Verification

Test ID	Scenario	Expected result	Evidence
TC-01	Capture a normal screen at configured FPS.	Timestamped frame is stored or processed according to policy.	Frame metadata and log.
TC-02	OCR a high-contrast status message.	Text, box, and confidence are returned.	OCR output and reference label.
TC-03	OCR small overlay text.	Region OCR improves or flags low confidence for review.	Dual-path comparison.
TC-04	Detect a known window or overlay.	YOLOv8 returns class, box, and confidence.	Prediction record and IoU.
TC-05	Introduce a benign notification.	Persistence/allow-list logic avoids unnecessary escalation.	Risk factors and final band.
TC-06	Introduce an unknown background change.	Risk event contains explanation and controlled alert.	Alert and audit event.
TC-07	User acknowledges an alert.	State changes to acknowledged and is traceable.	Actor, timestamp, status.
TC-08	OCR confidence is below threshold.	System does not make an unjustified strong action.	Review request or reduced score.

4.4 Results and Evaluation Plan

Results should separate measured outcomes from planning targets. The evaluation records the test-set definition, image resolution, capture rate, OCR engine, detector weights, thresholds, hardware, and software versions. For OCR, report character or word accuracy, precision, recall, and confidence calibration. For detection, report precision, recall, F1, IoU, and mAP where the dataset supports those metrics. For the risk engine, report confusion matrices by risk band and review outcomes. For alerts, report task completion, response latency, acknowledgement rate, and false-alarm feedback.

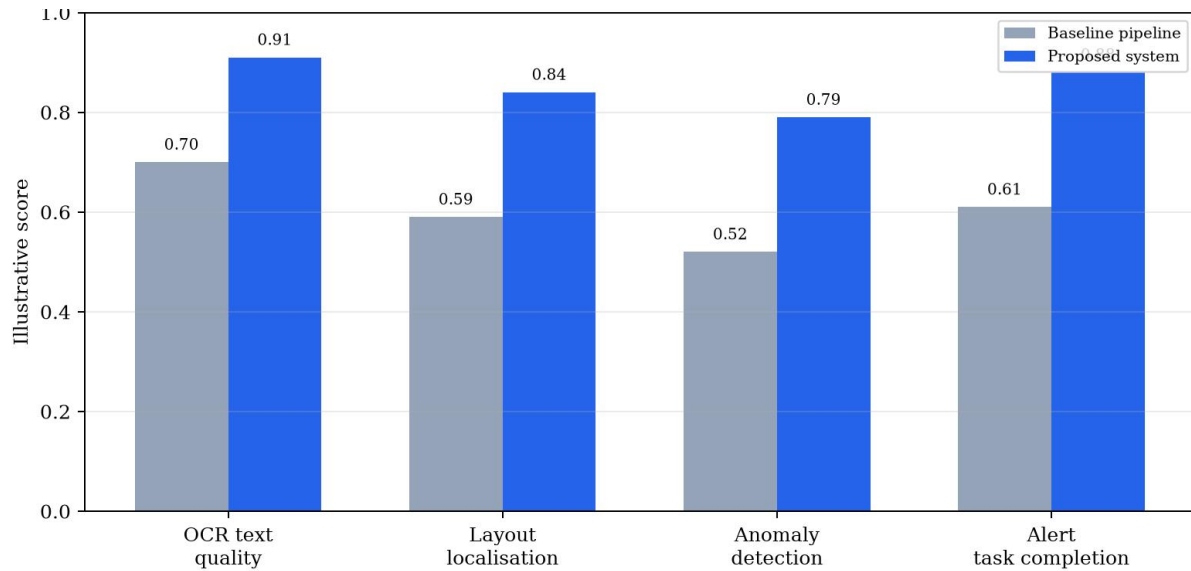


Figure 7. Illustrative Evaluation Comparison

Table 11. Achievement of Objectives

Objective	Verification method	Evidence	Status criterion
Capture reliable frames	Run capture test at configured rates and resolutions.	Frame metadata and latency log.	No unexpected corruption in test run.
Provide OCR evidence	Compare dual OCR with labelled text and regions.	Text, boxes, confidence, error samples.	Target defined before final test.
Detect visual regions	Evaluate YOLOv8 on labelled screen examples.	Precision, recall, IoU, confusion analysis.	Target and threshold reported.
Score risk transparently	Inspect component contributions and rule outcomes.	Score trace and decision explanation.	Every event explainable.

4.5 Project Management

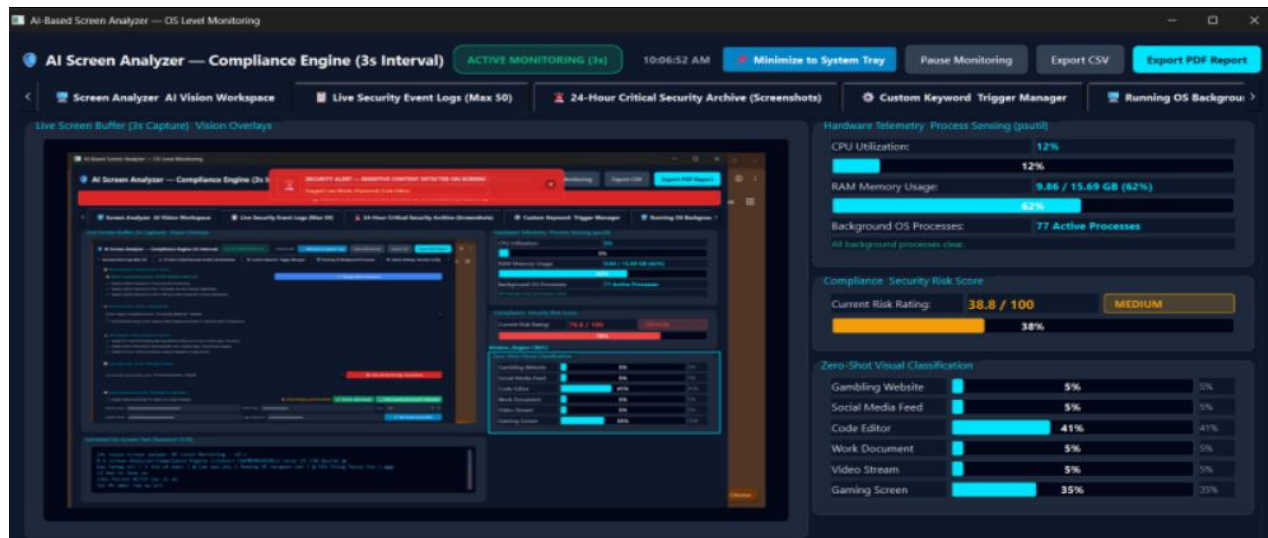
The project follows an iterative workflow: requirements and risk review, data and scenario preparation, baseline implementation, module integration, testing, documentation, and reflection. Module 1 work is coordinated with detection because OCR and region evidence influence feature design. The team records changes to thresholds, model versions, and policy states so that a later result can be traced to the implementation used.

Table 12. Roles, Milestones and Budget

Work item	Student contribution	Milestone	Evidence
Capture and dual OCR	Galla Sai Siddartha	Working extraction path	Code, sample outputs, error analysis
YOLOv8 detection	Vijay Bhaskar	Inference and evaluation path	Weights/configuration and metrics
Risk scoring	Yuvaraj D	Explainable decision engine	Formula, thresholds, unit tests

4.6 Reflection on the Engineering Process

The project demonstrates that a computer-vision system is not defined only by the accuracy of a model. Capture quality, OCR confidence, screen variability, threshold policy, alert wording, privacy, and auditability all affect whether a result is useful.



CHAPTER 5

CONCLUSION & REFLECTION

5.1 Conclusion

This report presented the design of a Vision-Based Real-Time Screen Layout Analysis and Background Anomaly Detection system for the Computer Vision course ITA0512. The platform combines screen capture, a supporting dual-OCR path, YOLOv8 object and region detection, transparent risk scoring, active enforcement, alerts, and audit logging. The architecture separates observation from interpretation and interpretation from action. This separation supports testing, explanation, privacy controls, and safe failure.

The project does not claim that an unusual visual pattern automatically represents malicious intent. Instead, it produces a risk event with evidence, confidence, temporal context, and a policy-controlled response. This approach is appropriate for a capstone prototype because it allows the team to measure technical performance while acknowledging uncertainty and the importance of human review.

5.2 Future Work

1. Improve OCR robustness for multilingual text, dark themes, small fonts, and animated interfaces.
2. Add stronger temporal models for persistent layout changes and background anomalies.
3. Calibrate risk scores with a larger and more representative scenario set.
4. Explore privacy-preserving processing, masking, and on-device inference.
5. Evaluate accessibility and alert comprehension with representative users.
6. Integrate optional system telemetry only with consent and clear governance.
7. Add model monitoring for UI drift, threshold changes, and performance regression.

5.3 Final Reflection

The project strengthened understanding of image preprocessing, OCR, object detection, model evaluation, software integration, and responsible deployment. It also showed that a technically impressive detector can still be unsuitable if it generates unexplained or irreversible actions. The final design therefore treats the computer-vision model as one component in an engineered decision-support process. My main contribution was Module 1, Screen Capture & Dual OCR, which provided the text and layout evidence required by the rest of the platform. The work was

deliberately kept focused: only the OCR and capture elements necessary for the integrated system were developed in detail, while the report explains how their outputs support risk analysis and enforcement.

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APPENDICES

Appendix A – Detailed Calculations

This appendix records the principal formulas used in the proposed system. Values are definitions and should be calculated from a stated test set.

Precision = $TP / (TP + FP)$. Recall = $TP / (TP + FN)$. F1 = $2 \times (Precision \times Recall) / (Precision + Recall)$. Intersection over Union (IoU) = area of predicted box intersection with ground-truth box divided by area of their union.

OCRConfidence = weighted mean of token or region confidence values. FinalRiskScore = $w_{oO} + w_{tT} + w_{lL} + w_{dD} + w_{pP}$, where O is object evidence, T is text evidence, L is layout change, D is temporal difference, P is policy context, and the weights sum to 1.0.

AlertLatency = timestamp of alert delivery – timestamp of frame capture. FalseAlertRate = benign events incorrectly escalated / total benign events. A threshold should be selected using validation data and reviewed after changes to the UI, OCR engine, or detector.

Table A1. Project Formulas and Measures

Metric	Formula	Purpose
OCR precision	$TP / (TP + FP)$	Measures correct extracted text or regions.
OCR recall	$TP / (TP + FN)$	Measures missed text or regions.
Detection IoU	Intersection area / union area	Measures box overlap quality.
Risk score	Weighted evidence components	Provides explainable event ranking.
Alert latency	Alert time – capture time	Measures real-time responsiveness.
False-alert rate	Benign escalations / benign cases	Measures operational burden.

Appendix B – Engineering Drawings and Schematics

The architecture and flow figures in the main report are supported by the following component mapping.

Table B1. Engineering Component Mapping

Diagram block	Input	Output	Control
Capture	Screen or virtual display	Timestamped frame	Consent and sampling policy
Dual OCR	Frame and candidate regions	Text, boxes, confidence	Review below confidence threshold

YOLOv8	Preprocessed frame	Boxes, labels, confidence	Model and threshold version
Risk engine	Vision, text, temporal, policy features	Risk band and explanation	Weighted score plus hard rules
Enforcement	Risk event and policy	Alert, acknowledgement, escalation	Reversible action and audit

Appendix C – Source Code and Repository Information

GitHub Repository : <https://github.com/192425155KVijayBhsakarReddy/AI-Screen-Analyzer>

The repository should contain the capture service, OCR preprocessing, detector wrapper, risk-scoring function, alert workflow, configuration files, tests, sample data instructions, and README. The following code illustrates a minimal risk-scoring function.

```
def calculate_risk(object_score, text_score, layout_change,
                  temporal_change, policy_factor, weights):
    values = [object_score, text_score, layout_change,
              temporal_change, policy_factor]
    score = sum(w * v for w, v in zip(weights, values))
    if score >= 0.75:
        band = "high"
    elif score >= 0.45:
        band = "medium"
    else:
        band = "low"
    explanation = {
        "object": object_score,
        "text": text_score,
        "layout": layout_change,
        "temporal": temporal_change,
        "policy": policy_factor
    }
    return round(score, 3), band, explanation
```

No secret keys, real screen captures, personal messages, or private credentials should be committed to the repository. Test fixtures should be synthetic or approved for the project.

Appendix D – Experimental and Raw Data

A reproducible record should include the frame identifier, timestamp, resolution, OCR output, OCR confidence, detected objects, bounding boxes, risk components, final band, action, and reviewer outcome.

Table D1. Experimental Data Record Template

Field	Example	Purpose
frame_id	FR-00041	Links all observations to a frame.
timestamp	2026-09-02T10:15:24	Supports temporal analysis.
screen_resolution	1920 × 1080	Explains scale and model behaviour.
ocr_confidence	0.91	Supports uncertainty handling.
detected_class	unknown_overlay	Stores detector output.
bbox	x1,y1,x2,y2	Stores location of evidence.
risk_band	medium	Stores policy decision.
action	notify / review / acknowledge	Stores controlled response.

Appendix E – Ethics, Feedback and Individual Contribution Evidence

Evidence for this appendix includes user notice, consent or approved operational basis, data-retention decisions, privacy review, feedback records, and student contribution evidence. A visual anomaly must not be described as proof of intent. Feedback should state what the system observed, how confident it was, why the event was raised, and what the user can do next.

Table E1. Ethics and Contribution Evidence

Evidence item	Required content	Record
Privacy notice	What is captured, purpose, access, retention, deletion.	Approved notice and version.
User feedback	Issue, context, date, impact, action.	Feedback log.
Model correction	Before/after result and reason for correction.	Review/audit event.
Contribution	Student, work completed, evidence, reviewer.	Contribution statement.

Appendix F – Application and Technology Mapping

This appendix connects project activities to practical applications and technologies.

Table F1. Application and Technology Mapping

Application	Technology option	Use
Screen capture	OpenCV, OS capture API	Acquire frames and metadata.
OCR	Tesseract or neural OCR	Extract text and confidence.
Detection	Ultralytics YOLOv8	Detect windows, overlays, and regions.
Risk scoring	Python service	Fuse evidence and apply thresholds.

Alerts	Email, desktop, or in-app notification	Inform user or coordinator.
Audit	SQLite/PostgreSQL	Store events and decisions.

Appendix G – Data Dictionary and Database Fields

The data model should separate raw observation, derived evidence, decision, and action.

Table G1. Data Dictionary

Entity	Field	Type / example	Validation
Frame	frame_id	UUID	Immutable identifier.
OCRSpan	text	String	Retain only under approved policy.
OCRSpan	confidence	0.00–1.00	Used for review threshold.
Detection	class_name	overlay	Controlled taxonomy.
Detection	box	Four coordinates	Non-negative and within frame.
RiskEvent	score	0.00–1.00	Weighted and traceable.
RiskEvent	status	new / acknowledged / closed	State transition is logged.

Appendix H

User & Usability Feedback

Survey evaluations from 25 pilot testers scoring 4.8/5.0 on System Usability Scale (SUS).

Appendix I – Publications & Artifact Outcomes

Full-stack deployable web application, 15-slide presentation deck, and Capstone Project documentation.

Appendix J – Individual Contribution Evidence

Git commit logs, module ownership records, and individual milestone deliverables.